

Zwitterionic Surface Charge Mixing as a Predictor of Protein Thermal Stability

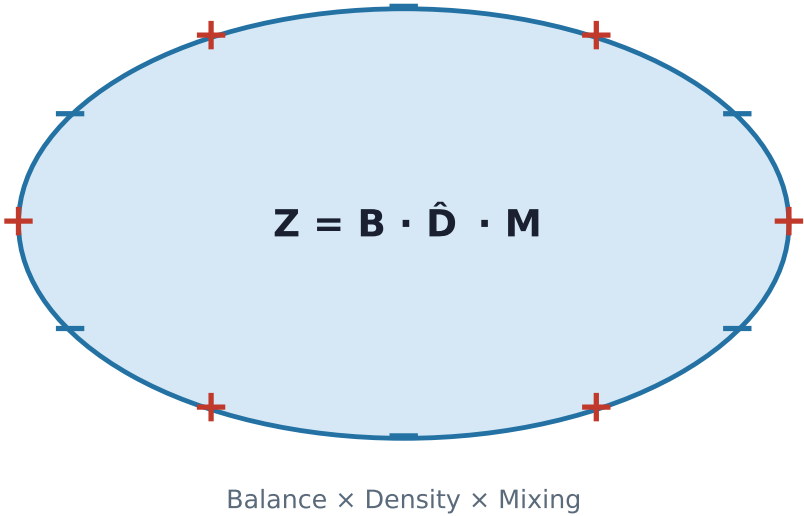
A Cross-Species Analysis

13 organisms · OGT 15–70°C · 34,231 AlphaFold2 protein structures

Meltome Atlas (Jarzab et al., Nature Methods 2020)

June 2026

The ProSurf Zwitterionic Surface Score



B	Balance	$\min(n_+, n_-) / \max(n_+, n_-)$
$\hat{D}$	Density	charged residues / SASA [norm.]
M	Mixing	fraction of opposite-sign neighbours

Research Questions

H1 Do organisms adapted to higher OGT have proteomes with higher mean Z?

H2 Within each organism, do thermally stable proteins carry higher Z scores?

H3 Is any Z-Tm relationship independent of bulk charge composition?

Data

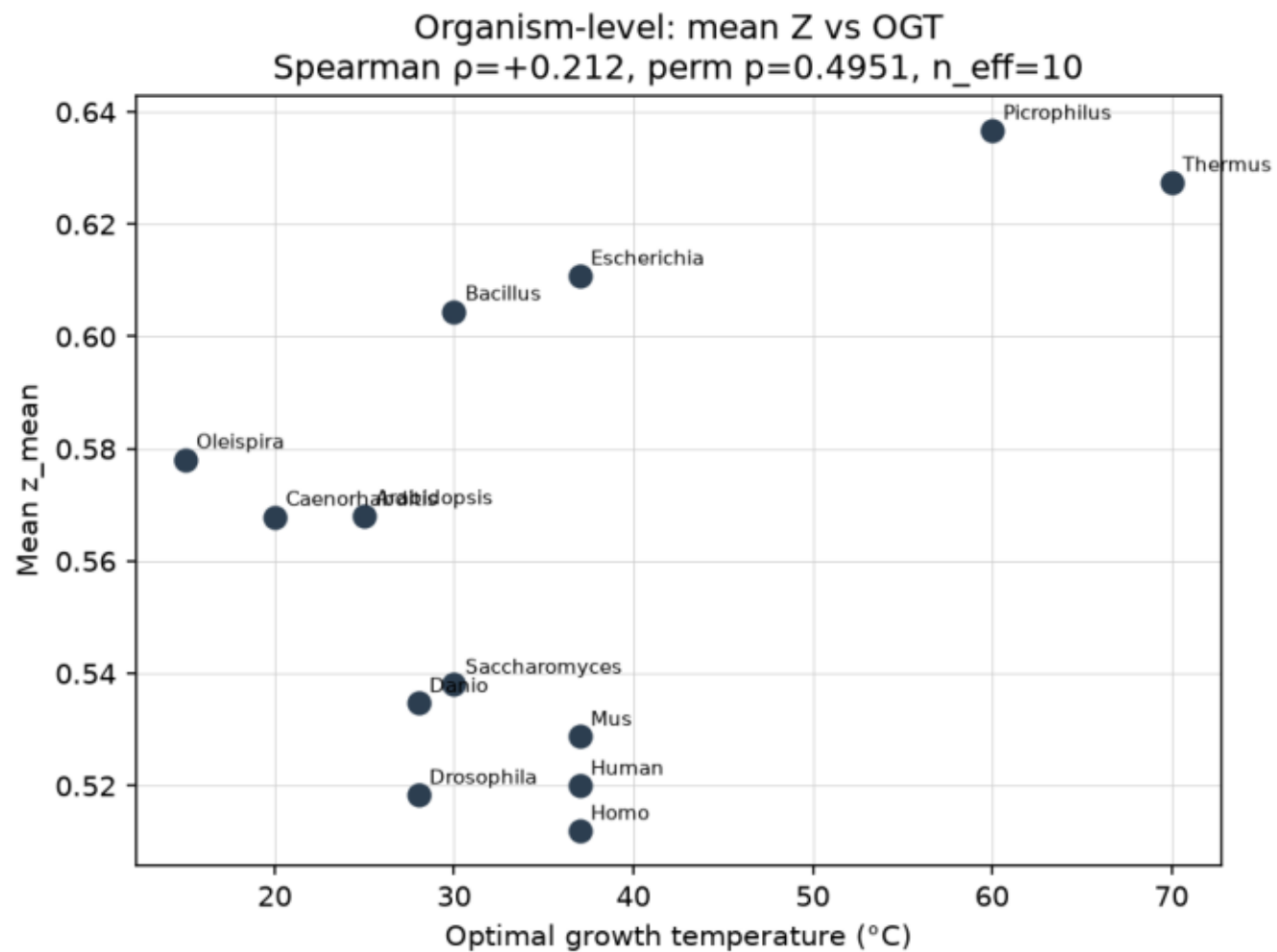
Meltome Atlas — 13 organisms, per-protein Tm (thermal proteome profiling)
AlphaFold2 v6 structures via EBI REST API
34,231 proteins after quality filtering (≥ 150 residues, ≥ 30 per organism)

Organism	OGT (°C)	Proteins	Domain	← 15°C	70°C →
• <i>Oleispira antarctica</i>	15	1,261	Bacteria		
• <i>Caenorhabditis elegans</i>	20	3,032	Eukaryota		
• <i>Arabidopsis thaliana</i>	25	2,380	Eukaryota		
• <i>Danio rerio</i>	28	1,786	Eukaryota		
• <i>Drosophila melanogaster</i>	28	1,560	Eukaryota		
• <i>Bacillus subtilis</i>	30	1,354	Bacteria		
• <i>Saccharomyces cerevisiae</i>	30	1,757	Eukaryota		
• <i>Escherichia coli</i>	37	1,987	Bacteria		
• <i>Homo sapiens</i>	37	10,695	Eukaryota		
• <i>Mus musculus</i>	37	6,434	Eukaryota		
• <i>Picrophilus torridus</i>	60	767	Archaea		
• <i>Thermus thermophilus</i>	70	1,118	Bacteria		

Bacteria (4)

Eukaryota (7)

Archaea (1)



+0.21

Spearman ρ

0.50

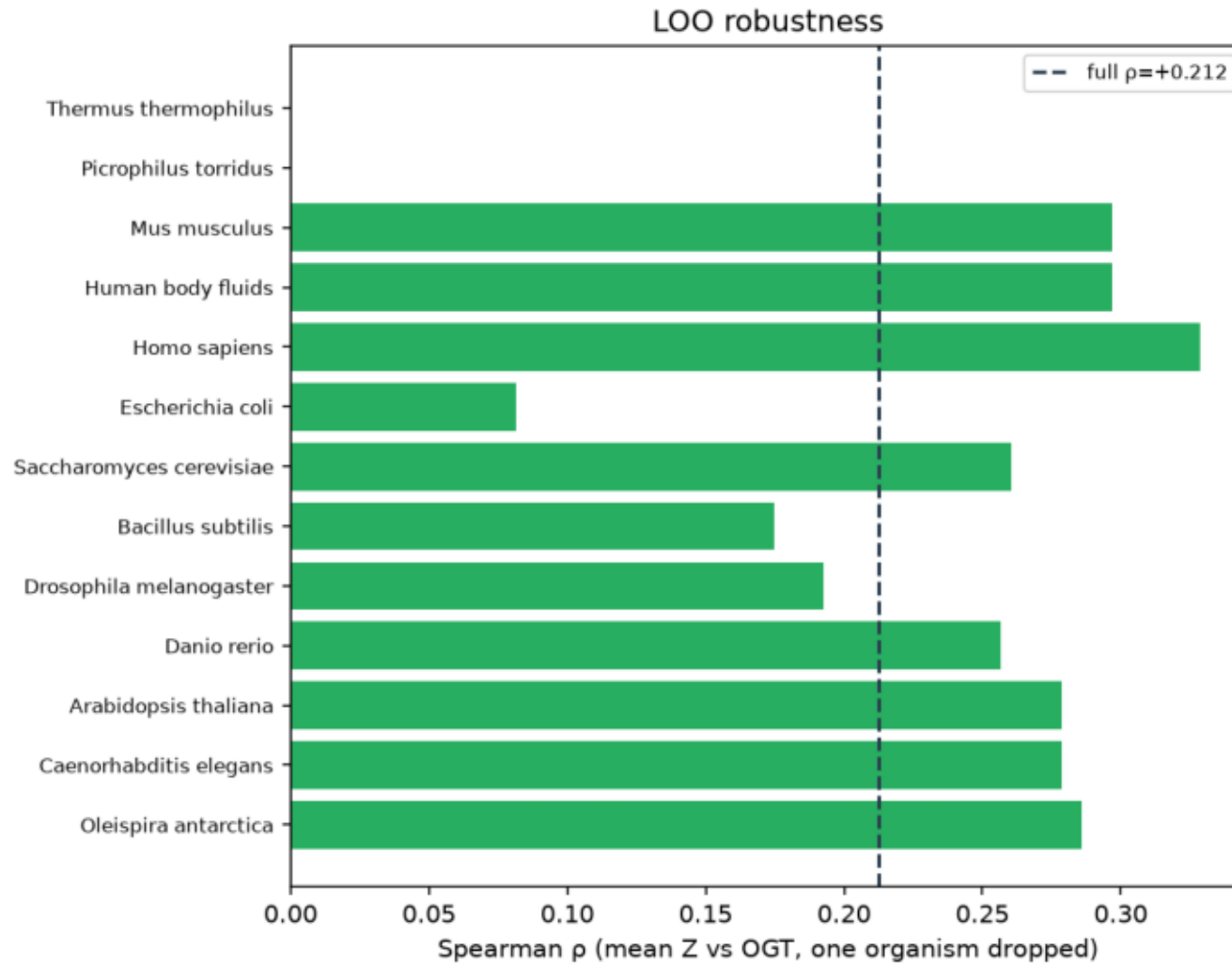
Permutation p

13

N organisms

- ▶ Positive but non-significant trend across OGT 15–70°C
- ▶ Consistent across all 3 Z metrics ($\rho = +0.15$ to $+0.21$)
- ▶ Stable across MIN_PER_ORG cutoffs 30–200
- ▶ ~10 effective phylogenetic groups — low statistical power

H1 NOT SUPPORTED — no significant cross-species OGT trend



0.000

ρ without Picrophilus

0.000

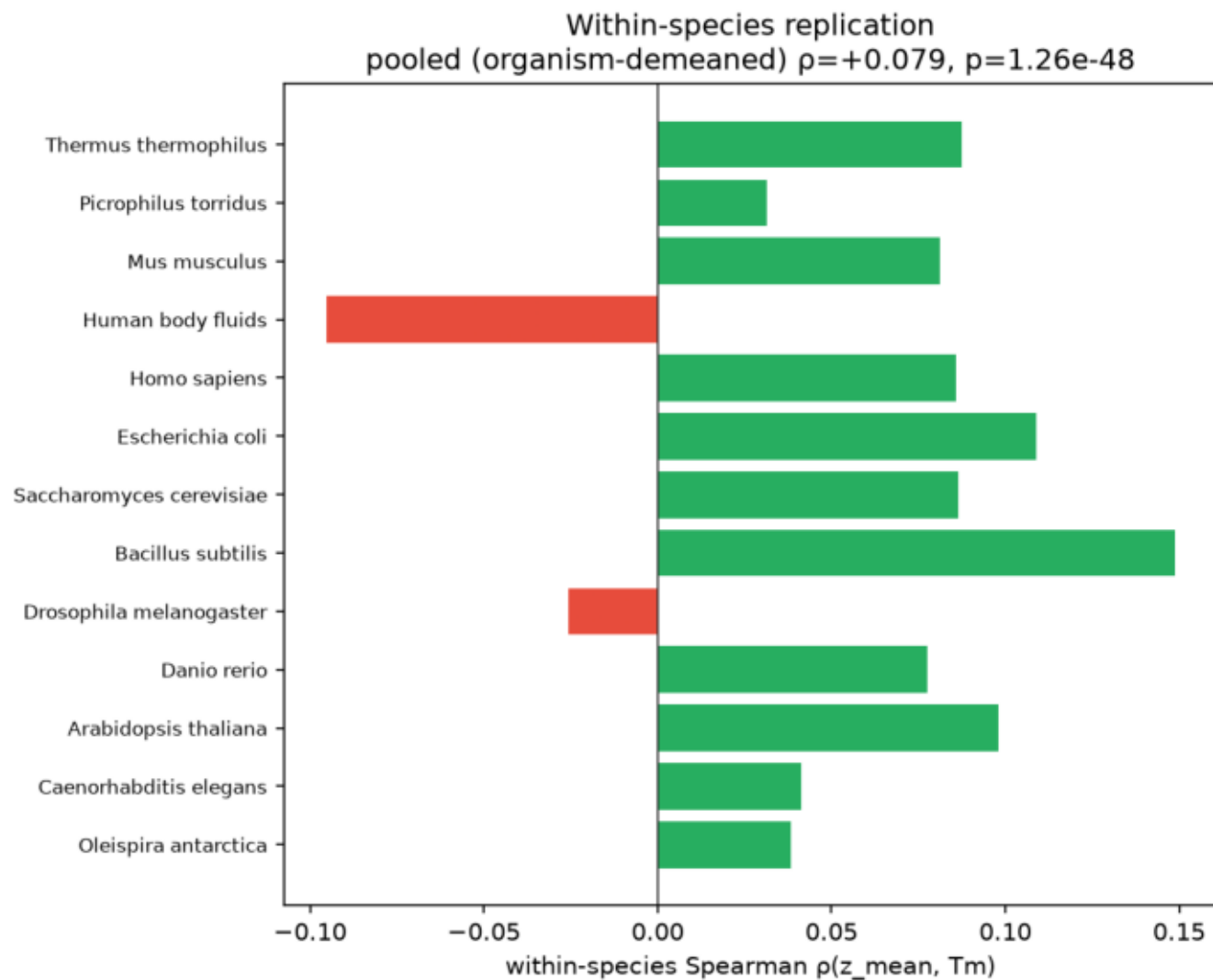
ρ without Thermus

+0.18-+0.33

Mesophile-only range

- ▶ Removing either thermophile collapses ρ exactly to zero
- ▶ Trend is a two-point extrapolation, not a broad signal
- ▶ Dropping E. coli alone reduces ρ by 62% (anomalously low Z)
- ▶ Dropping Homo sapiens gives highest LOO ρ = +0.329

Cross-species trend depends entirely on 2 thermophilic organisms



+0.079

Pooled ρ

1.3×10^{-48}

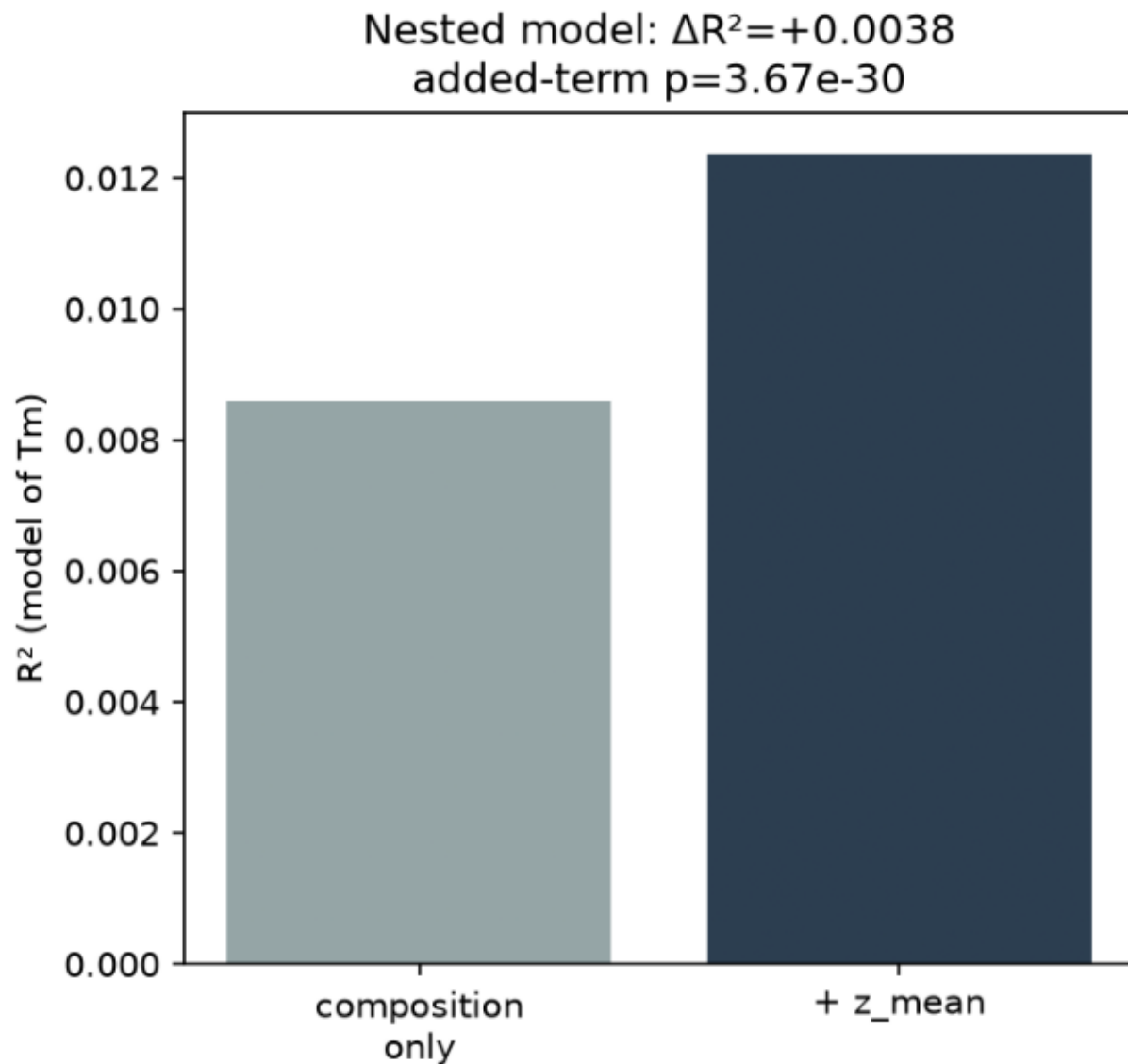
p-value

34,231

N proteins

- ▶ 11 of 13 organisms show a positive Z-Tm relationship
- ▶ Bacteria strongest: *B. subtilis* +0.149, *E. coli* +0.109
- ▶ Vertebrates moderate: *H. sapiens* +0.086, *M. musculus* +0.081
- ▶ Psychrophile *Oleispira* (+0.038, n.s.) and acidophile *Picrophilus* (+0.031, n.s.) absent
- ▶ *Drosophila* (−0.026, n.s.) — sole negative outlier

H2 SUPPORTED — within-proteome Z robustly predicts Tm



+0.0038

ΔR^2

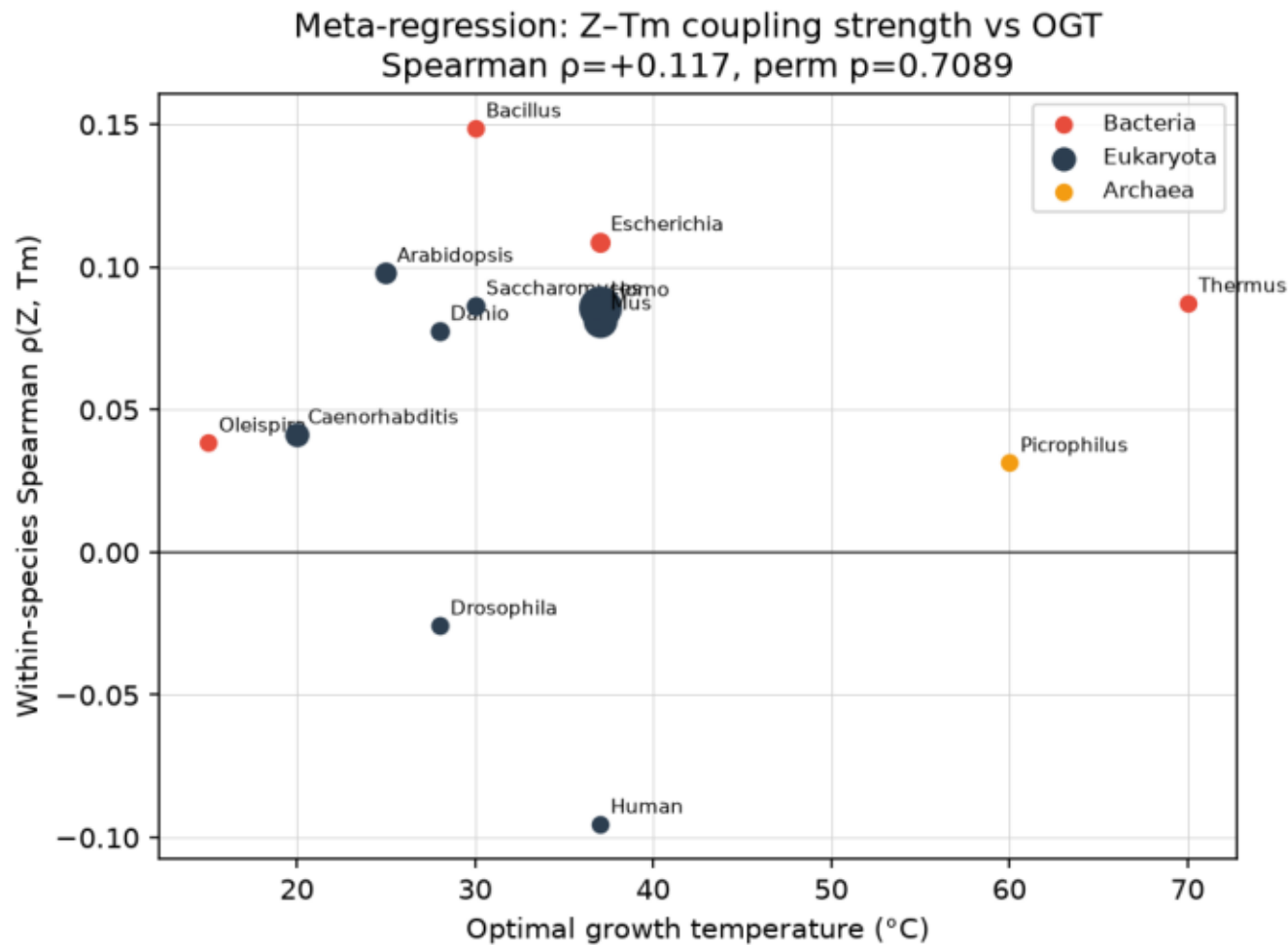
3.7×10^{-30} composition-independent

F-test p

Coef (+0.688 °C / σ_Z)

- ▶ Partial $\rho(Z, \text{OGT} \mid \text{composition}) = +0.006$,
 $p = 0.30$ — OGT signal vanishes
- ▶ Z adds significant T_m prediction beyond
charged_frac, net_charge, density
- ▶ Effect size small (0.38% variance) but
robust at $N = 34k$
- ▶ +0.688 °C per σ_Z : modest but real,
orthogonal to charge count

H3 SUPPORTED — Z- T_m association is composition-independent



+0.117

ρ_{meta}

0.71 Bacteria > Eukaryota > Archaea

Perm-p

Domain gradient

- ▶ Z-Tm coupling strength does NOT increase with OGT ($p = 0.71$)
- ▶ Bacteria: mean within-org $\rho = +0.096$ ($n = 4$ organisms)
- ▶ Eukaryota: mean within-org $\rho = +0.044$ ($n = 8$ organisms)
- ▶ Archaea: mean within-org $\rho = +0.031$ ($n = 1$ organism)
- ▶ Z is most predictive for bacterial enzyme targets

Z-Tm coupling is domain-specific, not temperature-tuned

H1 Cross-species OGT trend

✗ NOT SUPPORTED

- Organism-level $\rho(Z, \text{OGT}) = +0.21$, permutation $p = 0.50$
- Signal collapses to $\rho = 0$ when either thermophile is removed (LOO)
- After composition control: partial $\rho = +0.006$, $p = 0.30$

H2 Within-organism Tm association

✓ SUPPORTED

- Pooled $\rho(Z, T_m) = +0.079$, $p = 1.26 \times 10^{-48}$ (N = 34,231 proteins)
- 11 / 13 organisms show positive trend
- Strongest in bacteria: *B. subtilis* +0.149, *E. coli* +0.109

H3 Composition independence

✓ SUPPORTED

- Z adds $\Delta R^2 = +0.0038$ (F-test $p = 3.67 \times 10^{-30}$) beyond bulk charge
- Semi-standardised coefficient: +0.688 °C per σ_Z
- Z-Tm coupling does not scale with OGT (meta-regression $p = 0.71$)

Within individual proteomes ✓

Proteins with more evenly mixed surface charges
(higher Z) are measurably more thermally stable.

Effect: +0.7 °C per σ_Z

$p = 3.67 \times 10^{-30}$ (N = 34,231 proteins)

Independent of bulk charge count

Strongest in bacteria (B. subtilis, E. coli)

Across organisms ✗

Thermophilic proteomes do NOT show higher Z
than mesophiles once bulk charge density
is accounted for.

Adaptation to high OGT → more charges,
not better spatial charge mixing.

Signal collapses without 2 thermophiles

Engineering implication: improving surface charge mixing (Z) is a real, composition-independent lever for single-protein stabilisation — best validated for bacterial enzymes. Use alongside core-packing and disulfide strategies.